

Stability of Preference for Odors after Short-Term Exposure in Young Spiny Mice

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The stability of olfactory preferences for artificial odors was studied in young spiny mouse pups (*Acomys cahirinus*). Subjects aged between 2 and 20 days were exposed to the odor of either cinnamon or cumin for 1.5 hr. The durability of preferences for the familiar versus novel odor was subsequently monitored in a three-choice preference test. The results suggest the existence of a sensitive phase for learning odor characteristics, through simple exposure, between Days 2 and 18 of the pups' postnatal life. The sensitive phase proved not to be an "on-off" process, but the strongest effect of exposure to odors took place at about Days 4 and 6. However, the duration of preference for exposed odors was dependent on later experience with the odors during retests. The results point to an unusual plasticity in rapid learning of odors through simple exposure in precocial young spiny mice. © 1993 John Wiley & Sons, Inc.

Effects of exposure of newly born infants, usually of precocial species, to various kinds of species-atypical stimuli on the development of preferences for the stimuli have frequently been studied in mammals. These processes have usually been described as imprintinglike. For example, visual and auditory imprinting has been reported in guinea pigs *Cavia porcellus* (Harper, 1970; Sluckin, 1968). However, most mammals are macrosomatic and the olfactory system plays a crucial role in the development of social interactions in the young (Alberts, 1976; Leon, 1983; Rosenblatt, 1983) thus making olfactory stimuli more appropriate to study. Early olfactory experience with artificial odors has been shown to alter the olfactory preferences of guinea pigs (Carter, 1972; Carter & Marr, 1970) and in some ungulates to shift preferences to the scent of a different species which they had experienced early in life (Müller-Schwarze & Müller-Schwarze, 1971). Those experiments used a relatively long exposure to an experimental odor, during which

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Received for publication 14 September 1990

Revised for publication 14 August 1992

Accepted at Wiley 21 August 1992

Developmental Psychobiology 26(1):65–79 (1993)

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CCC 0012-1630/93/010065-15

that odor could have become associated with conventional reinforcers like food and warmth. In such cases associative learning might have been partly responsible for subjects' positive preferential response, or at least it could have interfered with the imprinting process, enhancing it considerably. Therefore, in the present study I chose the mode of simple brief exposure of young spiny mouse pups (*Acomys cahirinus*) to experimental odors in order to investigate further the olfactory imprinting reported by Porter and Etscorn (1976) and Janus (1989).

There are two criteria generally characterizing imprinting: existence of sensitive phases and relative stability of behaviors acquired within those phases (Immelmann & Suomi, 1981). The other two criteria, originally proposed by Lorenz (1935), namely, learning of species-species characteristics and termination of imprinting before the appropriate reaction is performed, should not be considered as general characteristics of filial imprinting. They apply to special types of imprinting like sexual imprinting, host selection, or to food or habitat imprinting. Therefore, in the present study I focused my attention on the existence and length of a sensitive phase, and on the relative stability of behaviors developed within that period.

Porter and Etscorn (1974, 1976) reported that olfactory preferences could be altered by a brief exposure in a precocial rodent species, the spiny mouse *Acomys cahirinus*. Spiny mouse pups exposed for 1 hr to an artificial odor (cinnamon or cumin) showed a preference for the exposed odor in a two-choice test 24 hr later. After one single exposure to cinnamon on the day of birth, spiny mice showed a significant preference for cinnamon over cumin on Days 3 and 6 (Porter & Etscorn, 1976). More recently, it has been shown that young spiny mouse pups exposed briefly either to cinnamon or cumin showed a significant preference for experimental odor 48 hr later in the first 16 days of the animals' postnatal life (Janus, 1989).

In the present study the olfactory sensitivity to young spiny mouse pups has been subjected to further detailed scrutiny. Two main problems concerning the development of olfactory responses in young spiny mice have been addressed. The first question addressed (Experiment 1) focused on long-term olfactory preferences resulting from exposure to artificial odors. The stability of preference was established through repeated testing after the subjects were exposed to the experimental odor. The necessary control for the effect of exposure to odors during retests was addressed in Experiment 2. The second problem of the study concerned the attempt to define with some precision a sensitivity length for rapid olfactory learning. The investigation of this problem was carried out in Experiment 3.

Experiment 1: Duration of Preference for Cinnamon and Cumin After a Brief Exposure Between Days 2–16

It has been shown that young spiny mouse pups exposed to either cinnamon or cumin between Days 2 and 16 postnatally show significant preference for the experimental odor 48 hr later (Janus, 1989). The present experiment examined the duration of preference for the odors of cinnamon or cumin experienced earlier. Subjects were tested longitudinally every 48 hr after the exposure to the experimental odor.

Method

Subjects

Subjects ($N = 235$, 128 litters) were *Acomys cahirinus* pups born in the laboratory breeding colony. Litter size ranged from 1–4 animals. Only the young of breeding pairs with previous parental experience were used in order to avoid any possible influence of the mother's primacy on the pups' behavior, as is known in rats (Wright & Bell, 1978). All breeding pairs were housed in individual plastic cages measuring $25 \times 32 \times 12$ cm and containing sawdust as bedding. Commercial rat and mouse chow (SDS Rat and Mouse No. 3 diet) and water were available ad libitum. Animals were provided with a supplement (apple and sunflower seeds) to their diet once a week. Breeding stock and experimental animals were housed at an ambient temperature of 22°C under a 12 : 12 hr dark/light reversed photoperiod (lights off at 0700 hr). Illumination was provided by four 100-W bulbs during the light phase and by four 15-W red bulbs during the dark phase. Breeding pairs were checked daily for the presence of newborn offspring. All pups were marked on the day of birth (Day 1) for individual identification. They then remained with their parents except for removal for experimental purposes and weekly cage cleaning.

Apparatus

The testing apparatus consisted of a circular arena 14 cm in diameter, surrounded by a wall 16 cm high. Three smaller circular enclosures, each 8 cm in diameter and 35 mm high, were attached to the wall of the arena at equal distances along the outside. The arena and the enclosures were made of white semitransparent Perspex, and the wall was made of aluminium and painted white. Animals placed individually in the center of the arena could enter each enclosure through a small entrance (3.5×3.5 cm) in the wall. The floor in each enclosure contained approximately 25 holes, each 2 mm in diameter, through which the odor could be investigated. The whole apparatus was sited within a further shallow enclosure.

Experimental odors were provided by sprinkling a small amount (approximately 0.01 g) of the appropriate odor substance on the bottom of circular dishes which were fitted tightly under the floor of the enclosure. The entire apparatus was thoroughly washed out with hot water and dried before each test and randomly rotated between tests to ensure that enclosures were not in the same position in relation to the source of light and the observer. Artificial odors selected as stimuli were commercially available ground cinnamon and cumin. The animals were exposed to a given odor in a specially prepared exposure cage of the same size as the breeding cage. It was fitted with a perforated false floor evenly spaced, (115 holes of 2 mm in diameter each) mounted 3 cm above the base of the cage. The cage had an airtight lid made of clear Perspex. An odor substance (approximately 2.5–3 g) was sprinkled evenly on the base of the cage. Such a procedure ensured that the exposure to the experimental odors during training and testing were made as similar as possible. Triplicate sets of exposure cages were used—one solely for cinnamon exposure, one for cumin, and one which was never used with any experimental odors for control groups.

Litters were randomly assigned to control, cinnamon- and cumin-exposed groups, and, within each group, to one of the five exposure age groups. The age

at exposure was 2, 4, 6, 12, or 16 days of age. At the given age, the whole intact litter (without parents) was removed from their home cage and placed in the training cage which contained an appropriate experimental odor source, or in the case of control groups, in the exposure cage which never contained any experimental odor source. The total exposure period lasted 1.5 hr and consisted of three ½-hr exposures to the odor with 15-min breaks in between. During the breaks the animals were returned to their home cage. The number of pups in each of the six exposure ages varied from $n = 13$ to 15 (9–10 litters), for control groups; $n = 14$ to 15 (5–9 litters) for cinnamon-exposed groups; and $n = 17$ to 22 (9–10 litters) for cumin-exposed groups.

The first odor preference testing was carried out 48 hr after the exposure to the odor stimulus for each group. Two hr before that test, each animal was given a 10-min habituation session in the apparatus. At that time none of the three enclosures contained any experimental odor source. During odor preference tests, the animal was given a simultaneous choice between cinnamon, cumin, and an enclosure which did not contain any artificial experimental odor (the bowl under the floor of that enclosure was empty). Each subject was given two 10-min tests with experimental odors, separated from each other by at least ½ hr. The stability of odor preference was established by retesting the animals every 48 hr after the exposure to the odor stimulus. Tests of the experimental groups (cinnamon- and cumin-exposed) were repeated until the animals' response to the familiar experimental odor did not significantly differ from the response to that odor of the animals from the control groups appropriate for each exposure groups. The number of retests for each group at exposure was determined in pilot experiments. For each pup, the total time spent investigating each enclosure was recorded. The animal was scored as investigating an enclosure when at least its head and forepaws were inside the enclosure. All tests were carried out in a separate room adjacent to the colony room, between 0900 and 1200 hr, that is, during the first part of dark phase of the photoperiod. The conditions in the testing room were the same as in the colony room.

Statistical Analysis

Litter-mean values were used as the units of analysis (Abbey & Howard, 1973). The cumulative amount of time spent within an enclosure was expressed as a proportion of time the animal spent in all enclosures. The data were then expressed as average proportions of time the animal spent in all enclosures during the two tests. These proportions were submitted to an arcsine transformation in order to normalize the distribution (Zar, 1984). Data were analyzed using a Group $\times$ Days ANOVA with retesting days as repeated measures (Winer, 1962). Such analyses were run for each group at each exposure separately. Separate planned comparisons (Sokal & Rohlf, 1981) between responses of experimental and control animals to an odor in question at each day of testing were carried out where appropriate by two sample t tests (two-tailed levels of significance). Whenever the variances were significantly heterogenous (Fmax-test) the degrees of freedom of the t tests were based on separate variance estimates (Sokal & Rohlf, 1981).

Table 1
Results of Two-Way ANOVA with Group (Odor-Exposed vs. Controls) and Days (Day of Exposure) as Repeated Measures for Each Experimental Odor.

ANOVA Factors	Age at Exposure (Days)				
	2	4	6	12	16
Cinnamon					
Group	17.69*	72.03*	50.98*	123.13*	31.80*
(df)	(1,16)	(1,13)	(1,12)	(1,14)	(1,15)
Days	11.08*	11.29*	7.34*	22.58*	74.20*
(df)	(2,32)	(7,91)	(6,72)	(3,42)	(1,15)
Group × Days	9.02*	16.37*	12.57*	28.51*	120.40*
(df)	(2,32)	(7,91)	(6,72)	(3,42)	(1,15)
Cumin					
Group	31.29*	76.55*	76.95*	135.95*	28.27*
(df)	(1,17)	(1,17)	(1,16)	(1,17)	(1,17)
Days	36.42*	16.34*	11.67*	29.70*	162.56*
(df)	(2,34)	(7,119)	(6,96)	(3,51)	(1,17)
Group × Days	32.50*	15.16*	11.02*	21.66*	162.63*
(df)	(2,34)	(7,119)	(6,96)	(3,51)	(1,17)

Degrees of freedom (df) are given in brackets.

* $p < 0.001$.

Results and Discussion

The first experiment showed that young spiny mice, exposed for 1.5 hr to an artificial odor (cinnamon or cumin) between 2 and 16 days of age, significantly increased the preference for the exposed odor in subsequent tests. However, the length of this preference varied with the age of exposure. Separate two-way ANOVAs (Group × Days interaction with days of retest as repeated measures) were calculated for each experimental odor at each age of exposure between odor-exposed and control groups. For all exposure groups ANOVA revealed significant effects of group, days, and group × days interaction, all levels of significance, $p < 0.001$. The results of ANOVAs for the groups exposed at different ages to cinnamon and cumin are given in Table 1.

Figure 1 shows the proportion of time spent in the enclosure with cinnamon by prior cinnamon-exposed and nonexposed, control pups. The longest preference for cinnamon was shown by the pups exposed to that odor at Day 4. This preference was significantly higher than the response of control pups until Day 14, thus lasting for 10 days. The duration of the preference for cinnamon of groups exposed at Days 6 and 12 was shorter and lasted for 8 and 4 days, respectively. Pups exposed at Days 2 or 16 showed a significant preference for cinnamon only for 2 days.

The results of the counter-balanced experiment in which groups of subjects were exposed to cumin at the same age as cinnamon-exposed groups are shown in Figure 2. Statistical significance of ANOVAs calculated for each age at exposure to cumin is given in Table 1. The analyses revealed significant effects of group, day, and group × day interaction, all $p < 0.001$. The statistical significance of preferences for cumin at each day of testing for each exposure age is given in

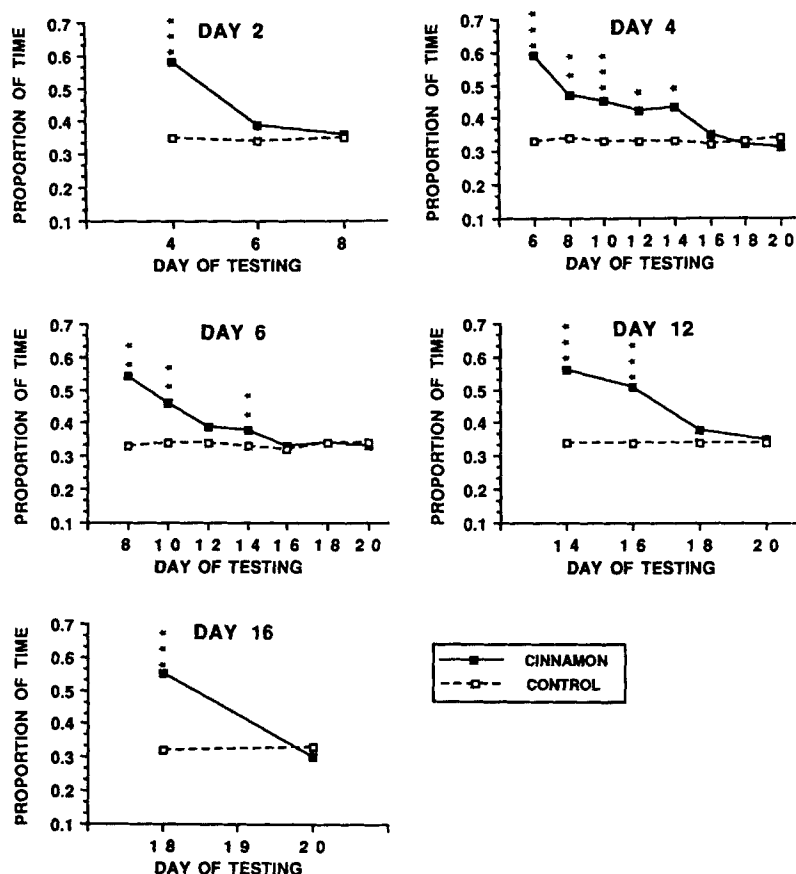


Fig. 1. Response to cinnamon of young spiny mouse pups exposed to the odor at Days 2, 4, 6, 12, and 16 postnatally. Each group at exposure and its appropriate control group were retested every 48 hr after the exposure. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed Student t test.

Figure 2. The pattern of responses to cumin by cumin-exposed subjects was similar to that observed in cinnamon-exposed groups. The longest significant preference for cumin (10 days) was shown by pups exposed to that odor at Day 4, while pups exposed at Day 6 showed preference only for 8 days. The pups exposed to cumin at Day 12 significantly preferred that odor for 6 days, and those exposed at days 2 or 16 showed significant preference for cumin for only 2 days.

The above results show that the stability of preferential response for an artificial odor experienced through a simple exposure depends very much on the age at exposure. For each experimental odor, the strongest positive response was established in pups exposed at age 4 or 6 days. The duration of preference in pups exposed at later ages was shorter, reaching 2 days only for subjects exposed at Day 16, which replicates my previous results (Janus, 1989). The 2-day-long preferential response to very young pups (exposed to the odor at Day 2) could be explained by their lower locomotor activity and lower level of arousal. These pups explored the exposure cage at much lower rates than older pups did, often sitting in a corner with eyes closed. It has been frequently suggested that a general

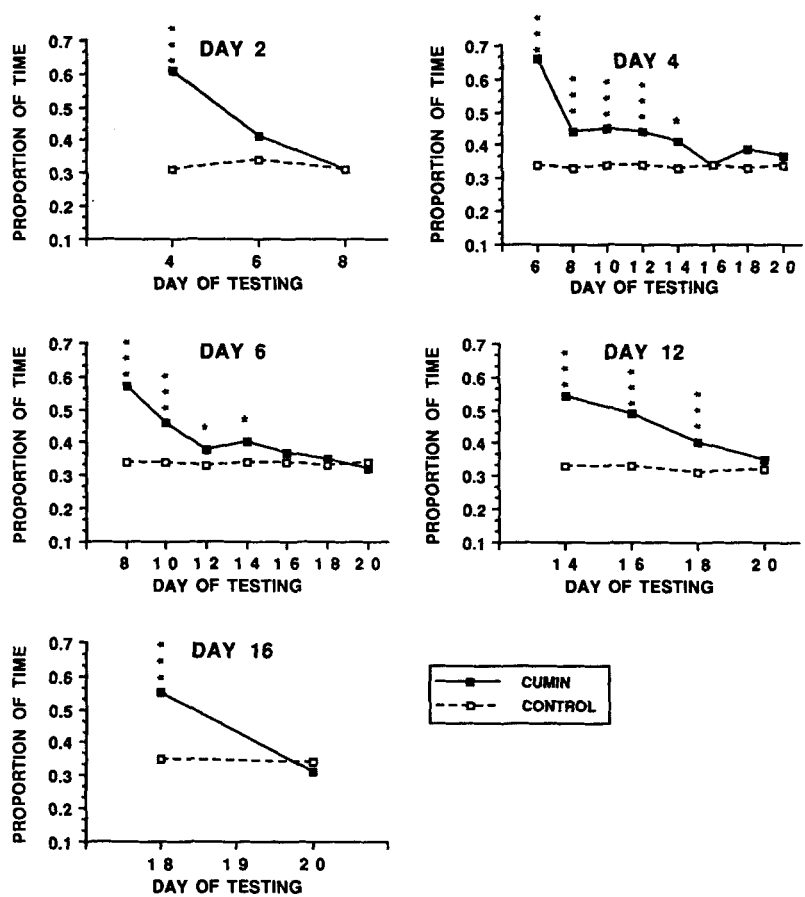


Fig. 2. Response to cumin of young spiny mouse pups exposed to the odor at Days 2, 4, 6, 12, and 16 postnatally. Each group at exposure and its appropriate control group were retested every 48 hr after the exposure. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed Student t test.

increase in the level of arousal is important in imprinting in birds (Fischer, 1970; Kovach, 1970; Martin & Schutz, 1974). The pattern of response to both experimental odors was very similar. The strongest response was observed during the first test after the exposure while the magnitude of preference in later retests, although similar, was maintained at a stable but lower level. The significant positive response for the exposed odor in repeated tests could, however, be maintained by exposure to the familiar odor in the testing apparatus. To control for this and other possible experimental artifacts, a set of tests was carried out as described in Experiment 2.

Experiment 2: Effects of Reexposure to Odors Occurring During Exposure and Repeated Testing

In this experiment, an attempt has been made to control for the effect of confounding variables which might exist during exposure to an experimental odor.

Two problems were addressed here. First, it is likely that the odor adhered to the subjects' fur and thus was present in the home cage during breaks between three 1/2-hr exposures. The positive response to the odor in later tests might therefore have resulted from reinforcing proprieties of the stimulus associated with littermates, parents, and a home cage. A second problem concerns the effects of reexposure to the odors during subsequent retesting after initial exposure. It is possible that such later reexposure to the familiar odor every 48 hrs, although very short, maintains the preference for it. A longer interval between retests should elucidate the problem of how the stability for an odor is influenced by reexposure to it. Because the age of exposure at 4 days produced the strongest positive response for the exposed experimental odor, subjects at that age were used for all tests in the present experiment.

Method

Two groups of pups at 4 days of age were exposed to cinnamon ($n = 13$, 7 litters) or cumin ($n = 13$, 7 litters). All the methodological procedures were the same as in Experiment 1 except that the subjects were not returned to the home cage between 1/2-hour exposures to odors but were kept in the holding cage instead. The results were compared by two-way ANOVAs with the results for Day 4 groups at exposure from Experiment 1.

In the second part of the experiment, Day 4 subjects were exposed to either cinnamon ($n = 29$, 11 litters), cumin ($n = 26$, 10 litters), or none of the experimental odors (control group; $n = 24$, 9 litters). Instead of retests every 48 hrs (2 days), they were given two subsequent retests every 4 days each, that is, Days 8 and 12 postnatally. The preference for cinnamon or cumin was compared with the response to appropriate odors of the control group.

Results and Discussion

The results of the control experiment examining the effects of littermates, parents, or the home cage on learning of the stimulus characteristics during breaks in the odor exposure revealed no significant effects of group and group $\times$ days interaction. The proportions of time for groups which were not returned to their home cages between exposure with the appropriate groups from Experiment 1 are shown in Figure 3a for cinnamon-exposed and Figure 3b for cumin-exposed groups. As predicted, there was a significant days effect. For the cinnamon-exposed group: group, $F(1,11) = 0.68$, ns, days $F(6,66) = 19.23$, $p < 0.001$, Group $\times$ Days, $F(6,66) = 0.33$, ns; for cumin-exposed groups: group, $F(1,15) = 0.26$, ns, days, $F(6,90) = 18.64$, $p < 0.001$, Group $\times$ Days, $F(6,90) = 0.28$, ns.

The results of the part controlling the effects of reexposure to odors during repeated tests did not reveal any significant effect of group, $F(1,18) = 0.08$, ns, days, $F(1,18) = 3.34$, ns, Group $\times$ Days interaction, $F(1,18) = 0.07$, ns, either for cinnamon-exposed or the cumin-exposed subjects, group, $F(1,17) = 1.06$, ns, days, $F(1,17) = 1.68$, ns, Group $\times$ Days, $F(1,17) = 1.25$, ns. The proportions of time of cinnamon- and cumin-exposed pups with their appropriate control groups are shown in Figure 4a and 4b, respectively.

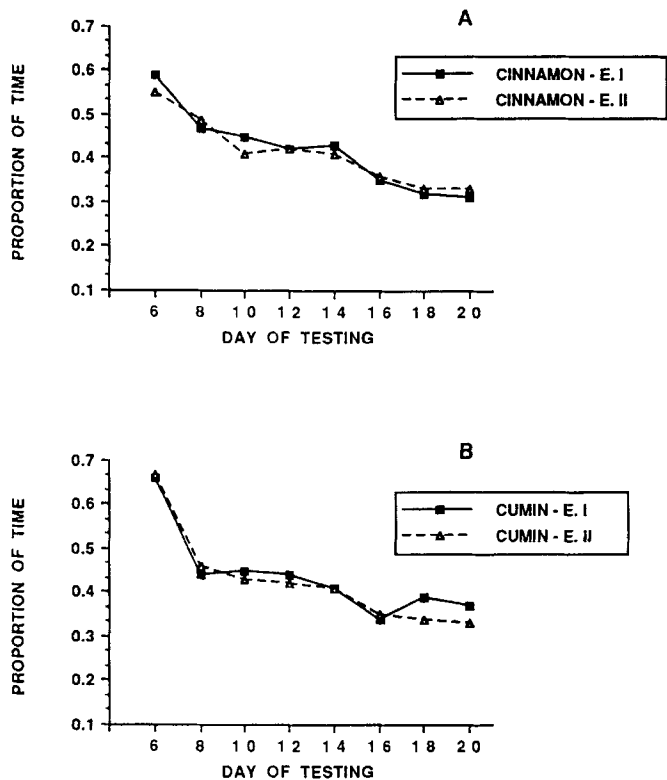


Fig. 3. Response to cinnamon (A) and cumin (B) of young spiny mouse pups exposed to the odor for three 1/2-hr sessions at Day 4 postnatally. Solid line represents the response of pups which were returned to their home cages during breaks in between exposures to odors (Experiment 1). Broken line represents the response of agemates which were not returned to home cages in between exposures (Experiment 2).

The obtained results clearly revealed that there was no association of the stimulus characteristics with parents, littermates, or the home cage during 15-min breaks between odor exposure sessions. Subjects that were not returned to their home cages during breaks did not differ in their preferences for the experimental odor from subjects of the same age that spent breaks in their home cages.

The results of the second part of the experiment revealed that subjects exposed to the experimental odor at Day 4 and repeatedly tested every 4 days did not show any significant positive response to that odor. Their response to the exposed odor did not differ from the response of controls that were not exposed to any experimental odor (group effect for both odors did not reach a significance level at $\alpha = 0.05$). The conclusion from the experiment is that the significant preference for the experimental odor is maintained by reexposure, albeit very short, to that odor during subsequent tests. A longer interval (4 days in the present experiment) between exposure to the odor and the first preference test does not produce significant positive response for the exposed odor. Moreover, a previous study (Janus, 1989) also showed that older animals, exposed to experimental odors at Day 18 and tested after the 2-day interval, did not show any significant preferential

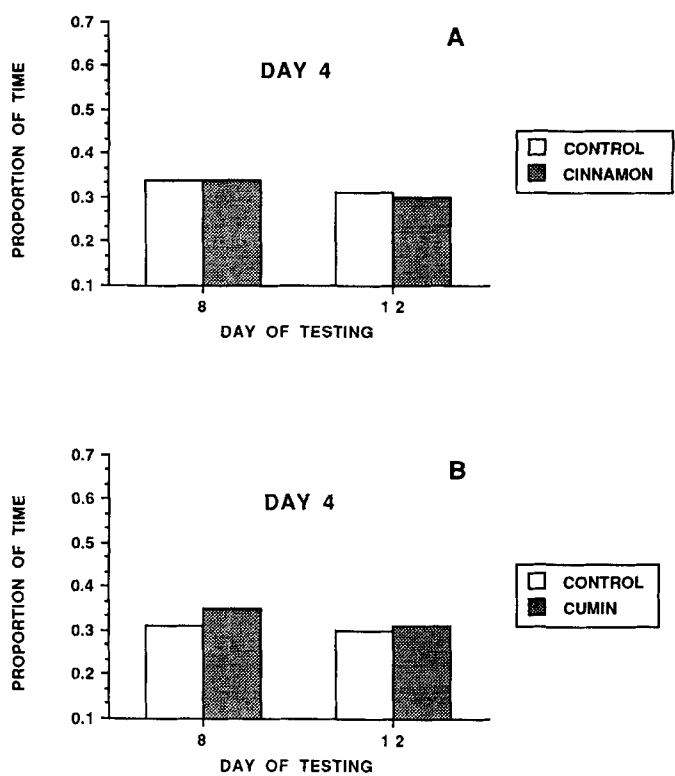


Fig. 4. Response to cinnamon (A) and cumin (B) of young spiny mouse pups exposed to the odor at Day 4 postnatally. Each group at exposure and its appropriate control group were retested every 4 days after exposure.

response for the experimental odor. Therefore the response of older animals to experimental odors was investigated in the next experiment (Experiment 3).

Experiment 3: Duration of Preference for Cinnamon and Cumin After Exposure Between Days 18–20

Although 18-day-old pups exposed to experimental odors failed to show any significant preference for them when tested 48 hr later (Janus, 1989), some individuals stayed longer in the enclosure with the familiar odor (personal observation). However, the possibility arises, that older pups do learn odor characteristics through a simple exposure, but the 48-hr interval between the exposure and the test is too long to reveal any preferential response. If shorter intervals between tests resulted in a preferential response for the exposed odor, the offset of a sensitive phase could take place later than 18 days, as suggested elsewhere (Janus, 1989). Thus, in the present experiment, subjects were exposed to cinnamon, cumin, or neither of the experimental odors at 18 or 20 days postnatally and retested after 24 hr.

Method

The number of pups in the exposure groups at Day 18 was $n = 14$ (8 litters) for the control group; $n = 14$ (5 litters) for the cinnamon-exposed group; and $n =$

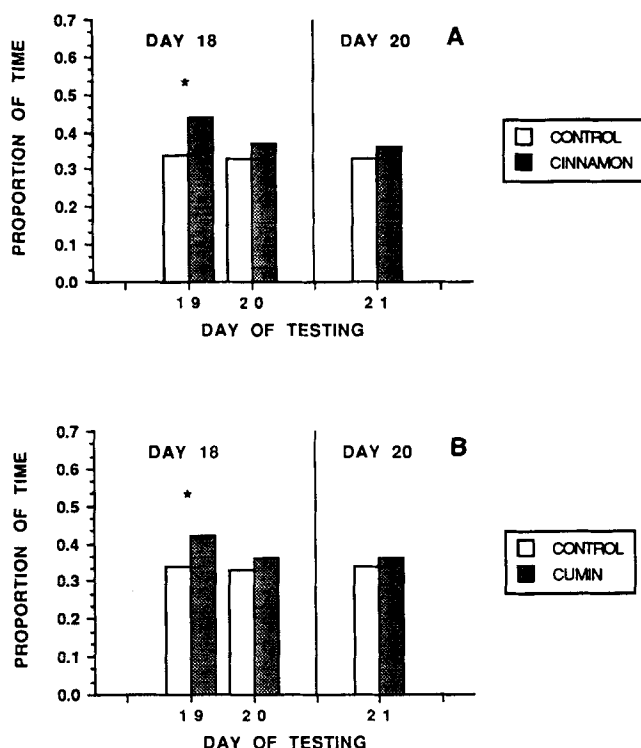


Fig. 5. Response to cinnamon (A) and cummin (B) of young spiny mouse pups exposed to the odor at Days 18 or 20 postnatally. Each group at exposure and its appropriate control group were retested every 24 hr after the exposure. * $p < 0.05$, two-tailed Student t test.

16 (7 litters) for cummin-exposed group. The number of pups in the Day-20 group at exposure was $n = 14$ (10 litters) for the control group; $n = 14$ (9 litters) for the cinnamon-exposed group; and $n = 14$ (9 litters) for the cummin-exposed group. All other methodological procedures were the same as in the Experiment 1.

Results and Discussion

Figure 5a shows the results of exposure of older pups aged 18 and 20 days to cinnamon and retesting after 24 hr. Only pups exposed at 18 days showed any preference in later tests: There was a significant effect of group, $F(1,11) = 16.60$, $p < 0.01$, and Group $\times$ Days interaction, $F(1,11) = 4.93$, $p < 0.05$. The significant preference for cinnamon occurred in a retest after 24 hr. The exposure of 20-day-old pups to cinnamon did not result in any significant preference for that odor in tests 24 hr later.

The results of a counterbalanced experiment with cummin are shown in Figure 5b. For the group exposed at Day 18 there was only a significant effect of group, $F(1,13) = 9.59$, $p < 0.001$. A significant preference for cummin occurred during the test 24 hr later. The exposure of 20-day-old pups to cummin did not result in any preference for that odor 24 hr later.

These results clearly show that the exposure of 18-day-old pups to experimental odors resulted in a preferential response for those odors. The response, however, lasted only 24 hr. The exposure to odors for 20-day-old pups, however, did not result in any significant preference for odors 24 hr later.

Discussion

The present results extend the findings of the previous study on the existence of a sensitive phase for the development of olfactory preferences in young spiny mice (Janus, 1989). They show that the sensitive phase for the development of preferences for odors experienced in the surrounding environment is much longer than the first 3 days of life reported by Porter and Etscorn (1976). The results of Experiment 3 also show that this olfactory sensitivity lasts 2 days longer than the previously reported 16 days of postnatal life (Janus, 1989), but the duration of acquired preferences at Day 18 does not last longer than 24 hr. Any exposure to the experimental odor at 20 days of life did not lead to a significant preference for it.

The combined results of Experiments 1 and 2 indicate that a brief simple exposure to arbitrarily chosen artificial odors of cinnamon or cumin resulted in a preferential response for the exposed odor within the first 18 days of postnatal life. However, the duration of this preference differed with the age at exposure. The longest duration occurred with pups exposed at 4 or 6 days of life. The duration of preference for the experimental odors for pups exposed at the age of 2 days lasted for 2 days, was maximal at age 4 and 6 days of exposure, and the duration of this preference decreased with age of exposure over 6 days so that in a group exposed at 18 days, it lasted only 24 hr. The animals exposed at 20 days did not show a preferential response to the odor, which suggests the end of the sensitive phase for the development of preference to nonbiological odors through a simple exposure.

The control for methodological procedure of exposure (Experiment 2) showed that the response of subjects to odors was not influenced or enhanced by the association of the exposed odor with reinforcing stimuli of the home cage.

Overall, the present results suggest that there is a sensitive phase for the development of preferences for artificial odors through a simple exposure which lasts until Day 18. It is not rapidly delimited "on-off" process because the strongest effect of such exposure takes place at about Days 4 and 6 of the animals' life.

The strong evidence for learning odors through a relatively brief simple exposure, together with the existence of a sensitive phase with a maximum effectiveness for the development of olfactory preference and its relatively long stability, might suggest a similarity with avian visual imprinting. The preference for odors in young spiny mice appeared after a brief period of exposure during which the odor was not associated with any conventional reinforcers. However, one cannot exclude the possibility that an association was established between the odor and the behavioral state of arousal caused by handling of the pups during the exposure session. As a consequence, the familiar odor could facilitate the exploration of enclosures and overall locomotor activity in preference tests. It has been shown that such an association can be established in young rats between a novel odor and a state of arousal caused by stroking in a 10-min session (Sullivan, Brake, Hofer, & Williams,

1986). When the rat pup encountered the odor in later tests, its ongoing behaviors were enhanced. On the other hand, in the present study spiny mice pups from controlled groups which were handled in the same way as the ones from odor-exposed groups but were not exposed to any experimental odor did not show any significant responses to any enclosure during the test at any exposure age. Their responses to odors were at the chance level (0.33, Figs. 1–3).

The results of the second part of the controlled Experiment 2 suggest that the observed relatively long preference for the exposed odor is maintained by reexposure to the odor in later tests. When the longer (4 days) between-test interval was used, subjects did not show any significant preference for the experimental odor experienced earlier. Thus, although the results confirm the existence of a sensitive phase for the development of preference for artificial odors through simple exposure, they also indicate that the duration of this preference depends on later experience, however brief, with the experimental odors.

The dependence of the duration of the olfactory preference, acquired within a sensitive phase, on later encounters with the familiar odor makes sense. It might help the animal to select relevant from irrelevant information about the surrounding environment. The olfactory characteristics of the habitat can change frequently. So can olfactory characteristics of conspecifics which might be dependent on the animals bacterial flora (Singh, Brown, & Roser, 1990), or on its diet (Porter & Doane, 1977). Thus, it seems likely that the olfactory “template” created within the sensitive phase must often be updated with relevant information while the redundant information is discarded. The dependence of preference duration on brief reexposures to familiar odors in later tests does not contradict the assumption of a relatively long preference for the stimulus characteristics learned during the sensitive phase. The experience with familiar odors during later tests involves the exposure of a pup to a stimulus without explicit pairing of neutral and significant events. Moreover, during retests the animal is exposed to the unfamiliar odor and also explores the “neutral” enclosure as well as the arena. Thus, this cumulative reexposure to a familiar odor is split into small bouts between which the animal experiences other olfactory cues. Therefore, it can be said that within a sensitive phase, learning of odor characteristics by simple exposure which is not associated with any reinforcers does indeed resemble the phenomenon of imprinting in birds. However, for mobile precocial spiny mouse pups, although it is important to learn the characteristics of an odor very quickly, it would be more adaptive if the duration of acquired preference for it depended on the frequency of later encounters with the odor. In those later olfactory experiences, however, the associative learning is the most likely process to dominate.

Another point which must be made clear is that the pups in the present study were constantly exposed to naturally occurring odors when in the home cage between the two 10-min tests with experimental odors. It is somewhat surprising that a brief simple exposure and later an even shorter reexposure (every 48 hr) to artificial nonbiological odors results in a relatively long duration of preferential response. The experimental design used is rather artificial, although the employment of a simultaneous multichoice test was an attempt to create a more realistic testing situation. It seems likely that odors present in the surrounding environment are experienced repeatedly over a longer period than the one used in the present study and frequently contain traces of odors from other conspecifics, thus often

having a biological significance for the animals. In this respect the present experiment assesses rather a plasticity (Miller, 1981) for rapid imprintinglike learning of odor characteristics by young spiny mice. Despite the lack of behavioral data from the field there is some evidence from laboratory studies which suggest a possible biological significance for the rapid development of preferences for odors through simple exposures. Spiny mice pups are born in a very advanced state of development, with their senses functional at birth and eyes opening shortly afterward (Ewer, 1968; Ruch, 1967). The newly born young are dependent on the mother's milk for only the first 6 days of life (Dieterlan, 1962). They start eating solid food as early as Day 4 (Porter, Cavallaro, & Moore, 1980) and can be successfully weaned in the laboratory at the age of 6 days (Porter, Doane, & Cavallaro, 1978). The females of this species do not build nests, so the young are not confined to one area and can move around freely. Despite the precocial state of development the young spend most of their time during the first 5 days suckling and huddling together (Porter et al., 1980). They begin to explore their environment (home cage) actively at about 4–6 days of life (personal observation). Thus the rapid development of long-lasting preferences for odors present in the surrounding environment seems to correlate with the pups' increased independence in exploration. In the rocky desert habitat of spiny mice such behavior might be biologically important for young pups. The strongest preference develops between 4 and 6 days when the young begin independent exploration of the surrounding environment but are still probably dependent on maternal care and protection. The preference for a familiar odor would enable pups to find a "home" area, orienting their behavior through effects of familiarity on preference. This very rapid and relatively long-lasting development of preferences through a simple exposure can be viewed as environmental imprinting, as suggested by Hess (1973). This kind of environmental imprinting is probably limited to the preweaning age when pups still need some protection and help from parents and conspecifics. It seems also possible that having precocial young with a more "open program" of information acquisition about the surrounding environment was a precondition that enabled the spiny mouse to colonize harsh semidesert or desert environments. Prolonged sensitivity of learning and recognizing details of environment might be important for young pups and any change in the environment is followed automatically by a corresponding change in the relevant preference. And only characteristics of those stimuli which are met most frequently are learned and have biological significance for animals.

Notes

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